

Beyond University Rankings? Generating New Indicators on Universities by Linking Data in Open Platforms

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The need for new indicators on universities is growing enormously. Governments and decision makers at all levels are faced with the huge opportunities generated by the availability of new knowledge and information and, simultaneously, are pressed by tight budget constraints. University rankings, in particular, are attracting policy and media attention, but at the same time receive harsh methodological criticism. After summarizing the main criticisms of rankings, we describe 2 trends in the user requirements for indicators; namely, granularity and cross-referencing. We then suggest that a change in the paradigm of the design and production of indicators is needed. The traditional approach is one that not only leverages on the existing data but also suggests heavy investment to integrate existing databases and to build up tailored indicators. We show, based on the European universities case, how the *intelligent* integration of existing data may lead to an open-linked data platform which permits the construction of new indicators. The power of the approach derives from the ability to combine heterogeneous sources of data to generate indicators that address a variety of user requirements without the need to design indicators on a custom basis.

University Rankings and the Demand for New Indicators in Higher Education and Research

The construction of new indicators is a crucial activity in policy making. Governments use numbers to govern complex situations (Desrosières, 2008; Porter, 1995). Researchers in social sciences are increasingly requested to

conceptualize, design, and implement indicators for various social and policy goals. Indicators combine existing information in order to generate new information that serves specific purposes.

A well-known example is the construction of university rankings, which is an exercise that attracts a lot of policy and media attention despite the numerous methodological shortcomings. The original goal of these indicators was to inform potential users of universities worldwide (i.e., Chinese students and their families; governments supporting students abroad). The need for this information was met by making use of existing data on scientific publications (at institutional level, without accounting for disciplinary composition) and ignoring other aspects of the activities of universities. However, the limitations of data are important and should not be underestimated. Indeed, it has been rightly emphasized: “Data are not simply addenda or second-order artefacts; rather, they are the heart of much of the narrative literature, the protean stuff that allows for inference, interpretation, theory building, innovation, and invention” (Cronin, 2013, p. 435).

In this article, we summarize the literature on the limitations of university rankings to find out by difference the need for a novel approach. We then describe two broad trends in the user requirements for indicators’ development: granularity and cross-referencing.

Policy making almost invariably requires the use of custom-built indicators. For example, governments may want to compare the competitiveness of their country or region with peers or with the rest of the world, in particular scientific or technological fields, in order to support the decision to invest resources. If data are not already shaped this way, governments will commission a study of experts or consultants, who will first define the perimeter of comparison, then build the indicators, search for data, and examine the results. There is nothing wrong with this approach,

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except that: (a) it produces information that is not re-usable by others; (b) it produces information that is not re-usable over time because it becomes obsolete; and (c) it is expensive.

A variant of this theme is that policy makers realize they have data in their repositories already, but they are not integrated; therefore, they cannot be used to address cross-domain policy issues (e.g., data on higher education students are not integrated with data on international mobility flows). Thus policy makers will commission a dedicated effort to integrate existing databases for future use. Information technology (IT) specialists and consultants will almost invariably suggest that a new database be built, which will integrate the existing ones on the basis of a detailed design of the queries to be carried out on the constituent databases. Again, there is nothing wrong in this approach, except that: (a) it postpones the problem: The newly created database will require a continuous effort to be fed with dedicated queries; (b) users must rely on the specialist knowledge of database specialists; and (c) it is expensive.

We show that an *intelligent* integration of existing data in an open-linked data platform may permit the construction of new indicators without having to design the indicators on a custom basis.

Although the article develops examples drawn from the European experience, it can be generalized to other regions of the world. For example, in the United States, a federal initiative has started to build up an infrastructure on science funding and outputs (Star Metrics: <https://www.starmetrics.nih.gov>; see also Lane, 2010), while commercial initiatives on research outputs of universities are also available (e.g., <http://www.academicanalytics.com/>). As another example, in Brazil, the Lattes platform (<http://lattes.cnpq.br/>) offers open information on more than 4,000 research institutions and their outputs. More generally, the global movement towards the standardization of research information is progressing rapidly, at the level of individual researchers (ORCID: <http://orcid.org/>), funding agencies (CERIF: <http://www.eurocris.org>; CASRAI [www.casrai.org]) or research organizations (ISNI: www.isni.org; Ringgold: www.ringgold.com). These initiatives are strongly supported by international scientific associations (see, e.g., CODATA <http://www.codata.org> and the VIVO network of scientists: <http://www.vivoweb.org/>).

Beyond University Rankings? Addressing the Criticisms

University rankings are an interesting subject for information science and technology (S & T). Despite the fact that they are criticized from a methodological and academic point of view, they are largely covered by the media and are diffused among users and decision makers. First, because university rankings attract a lot of political and media attention, international institutions at the world level, specialized research centers, and associations of universities devote studies to the subject, raising concerns of potential mislead-

ing effects on policies (Estermann & Nokkala, 2009; Estermann, Nokkala, & Steinel, 2010; European University Association (EUA), 2012; Institute for Higher Education Policy, 2007, 2009; Marope, Wells, & Hazelkorn, 2013; Rauhvargers, 2011, 2013; Salmi, 2009; Salmi & Saroyan, 2007). Second, they elicit academic inquiry from a variety of perspectives. Methodological foundations of rankings have been criticized repeatedly in the literature (Buela-Casal, Gutiérrez-Martínez, Bermúdez-Sánchez, & Vadillo-Muñoz, 2007; Charon & Wauters, 2008; Florian, 2007; Harvey, 2008; Ioannidis et al., 2007; Kroth & Daniel, 2008; Molinari & Molinari, 2008; Salmi & Saroyan, 2007; Taylor & Braddock, 2007; Tierney, 2009; Usher & Savino, 2006). More technically, a number of authors have started to examine the formal properties of composite indicators, raising doubts on mathematical coherence and statistical robustness (Aguillo, Bar-Ilan, Levene, & Ortega, 2010; Bornmann, 2014; Bornmann, Lutz, & Daniel, 2013a; Bornmann, Moya Anegón, & Mutz, 2013; Chen & Liao, 2012; Daraio, Bonaccorsi, & Simar, 2015a; Docampo, 2012; Safón, 2013; Saisana, D'Hombres, & Saltelli, 2011). In particular, Saisana et al. (2011) show that rankings are robust in the top positions but are less reliable elsewhere, and for a certain number of universities the variability induced by changes in the construction of the composite indicator is so large that all existing rankings are meaningless. Finally, the impact of rankings has been so pervasive that several book-length studies are available (Dehon, Jacobs, & Vermandele, 2009; Erkkilä, 2013; Hazelkorn, 2011; Kehm & Bjørn, 2009; Shin, Toutkoushian, & Teichler, 2011).

We do not review this literature in detail here, but we try to classify the main threads of criticism in order to frame, by difference, the need for a novel approach.

The main criticisms fall in three areas: content of indicators, methodology, and implications (see Table 1). In another article (Daraio et al., 2015a) we have addressed content issues (labelled *Monodimensionality*, i.e., lack of data on education, *Input-output structure* and *Size and subject mix*) and methodological issues (labelled *Robustness*). In this article, we also try to consider the criticisms based on political/ideological arguments, which are reported under the heading *Impact and implications* in Table 1.

Another dimension of university activity, which is often neglected in university rankings, is their mission.

As observed by Bornmann (2014, p. 428): "Although there are a number of 'dos and don'ts' when designing rankings or ratings, there will probably never be one that will do justice to the heterogeneity of the institutions covered and able to produce a valid image of the performance of all institutions." Thus, the quest for a perfect composite indicator from which to derive a unique ranking is bound to fail. While this is undeniably true, there is certainly room for going beyond the state of the art. In this section, we claim that most of university rankings' weaknesses can be addressed by changing the basic methodology from a composite indicator approach, based only on output measures, to

TABLE 1. University rankings: main limitations.

Limitation category	Weaknesses	Description and references
<i>Content of indicators</i>	<i>Subject mix</i>	Do not take into account the disciplinary composition of universities (Buela-Casal et al., 2007; Williams & van Dyke, 2006).
	<i>Size</i>	Bias towards large universities (Daraio et al., 2015a).
	<i>Educational output</i>	Not inclusion of teaching output and quality of education dimension (Cremonini, Westerheijden, & Enders, 2008).
	<i>Input data</i>	Fail to account for the relation between inputs and outputs, and between outputs and social outcomes (Bonaccorsi & Daraio, 2014, 2008; Fumasoli & Huisman, 2013; Maassen & Potman, 1990).
<i>Aggregation of indicators</i>	<i>Third mission</i>	Do not take into account alternative missions of universities.
	<i>Statistical robustness</i>	The weights used for the aggregation of indicators are arbitrary and lack theoretical foundation (Brooks, 2005; Lukman et al., 2010; Provan & Abercromby, 2000).
	<i>Deterministic setting</i>	They assume a deterministic setting, while bibliometric indicators incorporate several sources of errors (authors identification, timing of publications, name of affiliations). Confidence intervals instead should be analyzed (Goldstein & Spigehalter, 1996).
<i>Impact and implications of rankings</i>	<i>Relationships universities-State</i>	Some authors (Ferlie, Musselin, & Andresani, 2008; Rhoades & Sporn, 2002) tried to trace back the origins of rankings to changes in the relations between universities and the State.
	<i>Neoliberal devices</i>	The underlying rationale of rankings is ultimately ideological and neoliberal (Amsler & Bolsmann, 2012; Deem & Brehony, 2005; Hazelkorn, 2009).
	<i>Legitimation of categorical inequalities</i>	They give legitimation and reinforce categorical inequalities, institutionalizing them (Cantwell & Taylor, 2013; Rawlings & Bourgeois, 2004).
	<i>Self-fulfilling prophecies</i>	The distinctions produced by rankings are increasingly important and taken for granted (Espeland & Sander, 2007; Griffith & Rask, 2007; Stake, 2006).

a comprehensive input-output approach made possible by the “intelligent” integration of existing data sets.

The question we address is: What are the abstract requirements that indicators should fulfil to address the criticisms raised against the university rankings? We derived a list of requirements from the literature on rankings. University rankings should have:

- Coverage of all relevant activities of universities (not only research and education, but also third mission activities);
- Consideration of differences in the input structure;
- A lack of systematic biases due to size and subject mix;
- Statistical robustness of comparison;
- Fairness of comparison.

The requirement of fairness of comparison summarizes the following conditions discussed in the literature: (a) the universities under comparison have a similar input structure; (b) the trade-off between outputs is explicitly recognized in the measurement; and (c) a higher ranking is associated with a higher performance, given the input structure and the trade-offs between outputs.

In fact, the differences among universities can be transformed into quantities only if we recognize that universities do not have the freedom to access relevant resources, but are constrained by legal, administrative, and public finance considerations.

Rankings make use of information on the outputs of university activities. They do not take into account the resources or inputs that are utilized to produce the outputs; or, in some limited cases, they normalize output data by a raw indicator of the size of the institution. It is indeed possible to

normalize and standardize output data, making comparisons among universities more realistic. Normalization, for example, by headcount staff or full-time equivalent staff, may deliver more careful comparisons, although sometimes it could introduce distortion induced by the heterogeneity of their sources. On the contrary, in an input–output framework (Bonaccorsi & Daraio, 2004), universities are considered as strategic units that combine the available resources in the best way (multiple inputs; e.g., human, capital and physical resources) to produce their outputs (multiple outputs; e.g., teaching, research, and third mission results), given the context in which they operate (Bonaccorsi & Daraio, 2007).

This means that universities can produce different mixes between research and teaching outputs and that producing more of one output—given the structure of inputs—may imply producing less of another output. Recognizing this situation means that once the structure of inputs is fixed, in order to compare universities among themselves, units need to be compared with similar mixes of outputs, or compared along one output, keeping the other constant.

In an input–output model, by applying a conditional directional distance approach, we can compare the performance of universities with different missions and subject mixes (see Daraio et al., 2015a, for additional information on the approach).

Furthermore, the analysis of the evolution of indicators would be extremely interesting over time. It is clear, however, that such analysis requires the comparability of indicators across countries and over time.

We suggest that a shift in methodology might address these criticisms. Interestingly, this will also meet what we see as trends in user requirements for S&T indicators in

policy making (see also European Parliamentary Research Service [EPRS], 2014). We discuss these trends in detail in the following two sections while referring the reader to the subsequent section for a description of the main ideas (concepts) of a data infrastructure, which could allow for the development of such indicators.

The Trend Towards Granularity of Indicators

There is an increased trend towards what we call *granularity* of indicators. New indicators are explicitly requested to allow various kinds of aggregation and disaggregation, preserving desirable statistical properties, in order to address new policy needs. Traditional indicators have been mostly developed at a high level of aggregation at a national level, and, more recently, with some limited disaggregation at a regional level. In addition, traditional indicators do not allow for much disaggregation by nature of actor or by discipline, as we will show. For the policy needs of democratic governments after the Second World War, the traditional Organisation for Economic Co-operation and Development (OECD) aggregate statistical apparatus was quite adequate.

We argue that new policy needs demand a much higher level of granularity, along three directions.

The first direction of increase in granularity is *territorial*. Traditionally, S&T policies have been the domain of States, whatever their constitutional structure (e.g., in Germany and Spain the funding of universities is mostly done at a regional level; in the United States the funding of research is mostly federal while States have a role in funding higher education, etc.). This centralization has been considerably reversed by a number of constitutional reforms, as well as a trend towards regionalization of innovation policies in most OECD countries (OECD, 2011a). This articulation of *loci* of policy-making follows a debate that has placed the geographic dimension at the core of public policies for research and innovation. In high tech- and science-based industries, innovation policy rests on a clear understanding of the strengths of the regional scientific systems, which are not easily detected using current statistics. In a granular approach, data would be available for each cell of a region–discipline matrix, allowing a fine-grained analysis of areas of specialization and excellence.

The second direction of the granularity trend is *institutional*. An important distinction in the public research system is between higher education institutions (among which are universities) and public research organizations (PROs). This distinction, which is recorded in official research and development (R&D) statistics at a national level, is then lost in the construction of indicators at lower levels of geographic and disciplinary aggregation. Governments tend to follow separate channels of funding for universities and PROs, in order to avoid disruptive political fights. At the same time, the debate on the relative strengths and weaknesses of these different institutional forms of public research is very hot. Governments increasingly face the challenge of estimating the differential impact of funding on

these subsystems, as well as the opportunities for integration and collaboration among the subsystems (OECD, 2011b). However, if a government decides to reconsider the historical allocation of the budget, it encounters difficulty in comparing the productivity of these organizations, because their mix of inputs and outputs are largely different. In contrast, in a granular approach, the analysis might be carried out by disaggregating by discipline (thus making the comparison more realistic) and considering the input structure (funding and staff), the publication outputs and the nonpublication outputs of the organizations (education and, possibly, the third mission) jointly.

The third direction of the granularity trend is *disciplinary*. Until a few decades ago, in most OECD countries the relative shares of funding among disciplines was not considered a highly controversial policy issue. The distinction between mission-oriented research and curiosity-driven research was considered appropriate (Ergas, 1987). Areas of science associated with mission-oriented research (e.g., space, supercomputing) and making use of complex instrumentation (e.g., particle physics, astrophysics) were funded separately on the government budget. Other areas of science associated with curiosity-driven research were funded on the basis of *ex ante* selection, with relative disciplinary shares following historical patterns and areas of specialization. By and large, funding systems in the steady state were following the disciplinary composition of the underlying research population, with slight variations at the margins. This picture has been challenged in the last three decades for a number of reasons, which we will report here. New broad disciplines, such as information sciences, materials sciences, and life sciences, were born in the mid-20th century, but exploded in the late 1970s and 1980s in terms of potential for discoveries and rates of production of new knowledge, which created pressure on funding systems. In the same period, in the United States there was a remarkable shift of priorities from space-related (and subsequently, defence-related) fields to life sciences, starting with the Cancer War launched by President Nixon in 1971. Finally, at the end of the 20th century, the policy agenda seemed to be dominated by concerns about climate change, sustainability, and environmental issues, so that many expectations have been placed upon sciences that integrate knowledge across many layers of reality (from the molecular level up to large living systems or energy systems). Thus, we see a resurgence of sciences that adopt a systemic approach, such as ocean sciences, climatology, ecology, and food sciences. In sum, governments understand that they cannot proceed with a steady state allocation of resources, but must build up some rationale for the disciplinary mix (Stewart, 1995).

Policy makers seem to be interested in the scientific and technological specializations at a national or regional level. In Europe, recent studies show how the scientific profiles of Member States differ with respect to participation in the Framework Program initiatives (Campbell et al., 2013). In the United States, the Science of Science and Innovation Policy initiative has produced detailed maps of specialization,

which also shows how the funding of research reinforces specialization (National Research Council, 2014).

We argue that new indicators must be consistent with a trend towards territorial, institutional, and disciplinary granularity. Although the need for better granularity is visible and understandable, the state of the art of indicators is not adequate.

The Trend Towards Cross-Referencing of Indicators

Another powerful trend that we see in policy requirements is towards *cross-referencing*, or the ability of indicators to be combined in meaningful ways, preserving their statistical properties. For example, having spent money on statistics in higher education and research, governments start to ask why the two sources of information cannot be meaningfully matched; or why information on patents of universities cannot be combined with information on publications of universities, unless a dedicated, and expensive, project is built in order to address this specific question.

There are several important reasons for the cross-referencing trend. One reason comes from the large amount of literature on research and innovation systems (Lundvall, 2010; Nelson, 1993). Using a variety of approaches and methodological options, this literature has shown that what matters for performance is not only the level of expenditure or the quality of individual actors, but also the *interaction* among actors, or the flows within the system. This explains the large interest created by studies on science–technology flows, university–business collaboration, author–inventors, mobility of researchers, international cooperation, impact of research on regional growth, and more.

In all these cases, what is at stake is a relation, or interaction, between systems or subsystems, each of which is equipped with its own array of indicators, which have developed in relative isolation for the purpose of separate policy issues. Thus, when governments ask how much interaction with business is generated by the funding of universities, they do not understand why this issue is not only difficult to address, but also definitely impossible with the state of the art of existing indicators.

Another reason is somewhat contingent, but powerful. Governments have to come to grips with the impact created by sources of information that are different from official statistics, such as university rankings. These sources use existing indicators to build up rankings that convey very powerful and clear messages to large audiences. Governments are often presented with the challenge of reconciling the information that comes from official statistics with the information conveyed by rankings. A striking case, largely discussed in the dedicated literature, is the reaction of some European governments, particularly France and Germany, to the relatively poor performance of their universities in Shanghai rankings. For example, Hazelkorn (2007, 2009, 2011) has interviewed decision makers and members of governments around Europe and has found a widespread impact

of university rankings in excess of their information content, while Kehm and Bjørn (2009) and Erkkilä (2013) have discussed the impact on university administrators and faculty.

The traditional approach to manage these issues is the design of the integration of existing databases, using techniques in database management for data integration, validation, and quality. This approach can no longer sustain the increase in sophistication of requirements for indicators. A change in the existing paradigm is needed.

Generating New Indicators from a Platform

The development of new technologies in database design, data analysis, quality, integration, visualization, and related fields—all areas in computer science that have seen impressive achievements in recent years—allows a conceptual jump in the search for a solution for the dilemma discussed earlier. What follows is an attempt to build new indicators in higher education and research that are consistent with the literature on ontology-based data management (Calvanese et al., 2010; Lenzerini, 2011; Poggi et al., 2008). This approach has been proposed in the field of S&T indicators by Daraio, Lenzerini, et al. (2015). We advocate the exploration and exploitation of this approach in the field of design of indicators on universities, as well as in the production of evidence-based support to complex policy making.

The first step is conceptualizing research and innovation systems as complex systems formed by more elementary *entities*. There are several possible ways of doing this, as witnessed in the literature. One distinction is between an actor-centered perspective, which defines elementary action units in the system, and a function-based perspective, which, in contrast, works more on structural relations and flows within the system. We will try to reconcile these two perspectives. A starting point for conceptualizing this issue is to think in terms of *actors*; for example, universities, non-university higher education institutions, PROs, firms, governments, funding agencies, government agencies, and nongovernmental organizations (NGOs). The large amount of literature on systems of innovation and the systematic mapping of European landscapes carried out in recent years (Lundvall, 2010) allows us to build up almost complete maps of actors.

The second step is to develop *concepts* associated with entities. The development of concepts may be helped by the consideration of users' requirements for indicators (see next section). Rather than following strict definitions, actors should be conceptualized in terms of a list of associated concepts. For example, a university is associated with concepts, such as students, professors, disciplines, location, publications, etc. Each of these concepts is, in turn, susceptible to further conceptualization. For example, students are associated with concepts, such as nationality, country of their last degree, gender, field of education, etc. For each concept we should carry out an extensive abstract analysis of the underlying definitions and of the relations with the other concepts. It is not important that definitions are formally similar, since the higher level concepts will be generated by variable

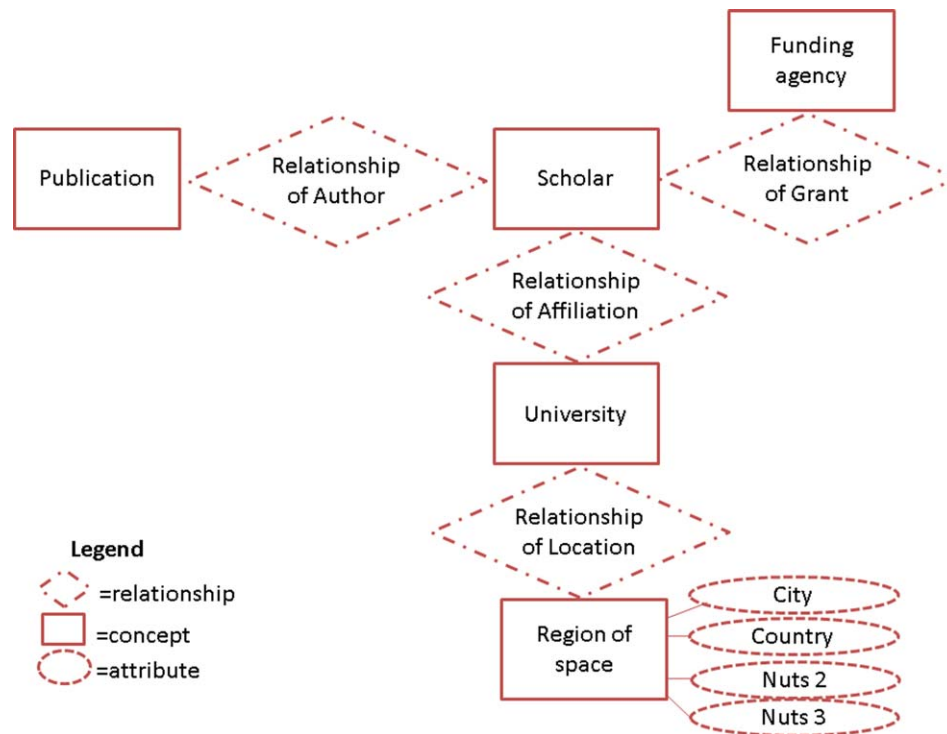


FIG. 1. An example of links between concepts. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/asi.23679)]

degrees of membership of the lower-level concepts. This is a crucial point. What is needed in the traditional integration of databases is a perfect matching between the concept and the definition in order to carry out a query. In this approach, instead, this issue does not prevent integration, as we will show. The similarity of definitions can formally be defined on the basis of the articulation of concepts. If two definitions are equal, their concepts overlap perfectly. If two definitions are different (for example, a “foreign student” might be considered a student of foreign nationality, or a student who has attended secondary school in a different country), then the associated concepts will reveal the difference. Given this approach, it is always possible to identify the subset of concepts that make two definitions similar. This abstract definition gives flexibility to the information system.

The third step is to look for lists and standards, or, more generally, in the database language, Authority Files. The existence of a list is enormously important, because it allows exhaustive searches. Lists are flexible tools, in the sense that they are associated with procedures for updating and correcting. For example, having an Authority File of companies investing in R&D, or a list of higher education institutions (HEIs) or of PROs, is an important step.

Standards are also extremely important. For example, one would take on board standard definitions adopted at an international level as to what constitutes an author of scientific publication (ORCID, see www.orcid.org), or a research organization (CERIF, see www.eurocris.org).

The fourth step is to establish links between concepts. For example, an important link between the concept of “university” and the concept of “scientific publication” is

via the concepts of “author” and of “affiliation.” Publications are conceptualized, by necessity, as produced by authors. Authors are conceptualized, by necessity, as associated with institutions (affiliation). The list of affiliations then has a non-empty overlap with the list of universities. Similarly, there is a link between “funding agency” and “university.” A funding agency, by necessity, funds research. A university, by necessity, carries out research based on funding received from someone else (i.e., universities do not self-finance). The link between these two concepts should then be articulated; for example, through the concept of a “grant.” A grant might be conceptualized as having a source and a beneficiary. The list of beneficiaries has then a non-empty overlap with the list of universities.

In Figure 1 we report an illustration of links between concepts. The basic elements are concepts represented as boxes; whereas the relationships are represented as diamonds and, finally, attributes are represented as circles attached to the concept for which they are defined.

The abstract notion of a link between concepts permits the incorporation of the functional perspective. In this perspective, flows describe the functions that actors of the system perform. For example, priority selection, or ex ante project evaluation, are necessary functions of a research funding system, whatever the concrete actor that will perform them. The functional perspective is useful because it offers an abstract view of the extremely complex landscape of actors. We incorporate this perspective using the even more abstract notion of “link between concepts.”

Once these steps are carried out, what we obtain is a graph of actors, each with an associated graph of concepts

TABLE 2. Generation of new indicators through the analysis of shared concepts.

Entity	Associated concepts	Shared concepts	Information needs	New indicators
University	• Location of site • Publications	a. Location is shared but only at higher level of abstraction: universities are single-site, PROs are multisite	Need to allocate publications of PROs at locational level, e.g., region (NUTS 2) or province (NUTS 3)	Number of publications per region/province
Public Research Organization	• Location of headquarters • Publications	b. Publications are shared		
University	• Location of site • Publications by discipline	a. Location is shared (see above)	Need to establish a standard for comparability of classification of disciplines	1. Number of publications per discipline per region/province
Public Research Organization	• Location of headquarters • Publications by discipline	b. Publications are shared		2. Map of scientific activity per discipline
		c. Discipline is shared but only at higher level of abstraction: need to examine lower levels of granularity of classifications		3. Scientific Specialization Index at region/province level (for example, Balassa index)
University	• Location of site • Publications	a. Location is shared (see above)		1. Publications per inhabitant at region/province level
Public Research Organization	• Location of headquarters • Publications	b. Publications are shared		2. Publications per euro of HERD
Location	• Inhabitants • HERD (Higher Education Research and Development)			

Source: Our own elaborations.

(direct and indirect link). The links between actors are represented as links between concepts associated with actors.

Remember that at this stage we have an abstract representation, for which the actual content of information might be variable. For example, we might have a perfect concept of “university,” but not have a list of universities at all. Or we might have a clear notion of “R&D performing firms,” but are not able to link their list to the list of patent assignees.

Then we are ready to implement a principle for the integration of research and innovation information systems. The informal principle we follow is: *Put your information in the region of the graph in which the potential for generation of other information is greater.*

For example, the concept of “location” is linked to the concept of “university” (all universities have a physical location in a place). But the concept of “location” is also related to a host of other concepts, such as “R&D expenditure” (it is possible to sum up all expenditures in a given region), “high-tech firms” (it is possible to count all firms in a region belonging to a category), or “patents” (patents have inventors whose location is registered).

This is not always so for other regions of the graph. For example, one might be interested in understanding the extension of patents to other countries. But the location of the countries in which firms or universities extend their patents is not conceptually linked (unless in very general terms) with the entities of the research and innovation system in Europe. Thus, this information would be extremely interesting, but weakly connected.

By using this approach, a number of new indicators can be generated. Table 2 illustrates the potential of the approach. We combine actors (or entities) in the graph by listing (some of) the concepts associated with them. When

two actors share the same concept, it is important to examine at which level of abstraction the overlapping takes place.

For example, universities and PROs share the concept “location” (they have a geographic location where their activity is carried out) but universities are, in most cases, single-site organizations (hence the location of the headquarters is the same of the departments or schools), whereas PROs are multisite organizations, so that the location of the headquarters does not provide information on the actual location of the research activities. By discovering this mismatch, it is clear that there is a need for investing in the production of new information to allocate research activities of PROs at a territorial level. Once this information is provided, the fact that universities and PROs share the concept of “location” at all relevant levels of abstraction implies that their publications—another concept they share—can be aggregated at the location level. From this observation, new indicators can be generated; for example, the number of publications at a provincial or a regional level.

The notion of location of a university should be qualified. It is true that universities can now engage in distance education, particularly using Massive Open Online Courses (MOOCs). However, there is no evidence that the virtualization of teaching has substantially altered the role of universities as engines of growth at both a local and regional level, particularly in research activities. It is, in fact, quite the contrary. Almost all recent studies confirm the general finding for universities that most of the impact is experienced in areas in which the benefits of proximity (i.e., face-to-face interaction) are still significant, typically not exceeding 100–200 kilometres from the location (Bonaccorsi, 2014; Boschma, 2005; Boschma & Frenken, 2010; Bottazzi & Peri, 2003).

Finally, the concept of “location” at a geographical level is associated with other concepts, such as population (or inhabitants), gross domestic product or R&D expenditure (e.g., gross domestic expenditure on R&D). All these associations of concepts come from a thorough conceptual analysis, which needs to be carried out before the construction of indicators. From the previous analysis, we have seen that both universities and PROs share the concept of location. Hence, all concepts associated with location can be linked to them. As an example, we can build up new indicators that relate the number of publications of universities to those of PROs at location level (province or region), which are then divided by the number of inhabitants, or divided by the monetary value of R&D expenditure. In this way, we build up new indicators that are made possible by the underlying shared concept structure; that is, location.

The advantage of this approach is that new indicators are built following the logical structure of entities and the internal hierarchy of concepts. In this way, entities that share concepts at the appropriate level of abstraction can be fruitfully combined. This, of course, does not mean that we are able to embrace reality in a hierarchical database (on the relationship between databases and reality, see Edwards, 2010; Edwards, Mayernik, Batcheller, Bowker, & Borgman, 2011).

The architecture of the platform goes beyond the hierarchical relational nature of traditional databases, since it permits the creation of links among concepts in a free way. All concepts may be related to any other concept, provided a rationale can be offered to interpret the new concept. On the other hand, the degree to which the concepts actually represent the “reality” will be open to criticisms and reformulations. Although the traditional “silos” approach to indicators makes it impossible for users to change the structure of data, the new approach is much more flexible and permits experimentation and critical analysis.

It should be noted that this approach is orthogonal to the current approach to indicators, which starts from user requirements (usually very demanding) and builds up an ad hoc integration of existing databases, without changing the conceptual structure of data. Rather, as stated at the beginning of this section, we advocate for an ontology-based data management approach (Lenzerini, 2011). This approach is based on a thorough conceptual analysis of concepts and their relations, after which the combination of data is greatly facilitated. In this way, the integration of data sets becomes a smooth process, because the underlying conceptual structure has been clarified *ex ante* (see Poggi et al., 2008). In our approach, once a new indicator is identified, there is guarantee that it will be feasible. If this is not true, we identify with great clarity the areas where additional investment in the creation of information is needed. This is a major step forward for the construction of indicators.

The construction of indicators from an open-data platform should not be left completely in the hands of users. In fact, in general, the statistical properties of new indicators are not necessarily consistent with the statistical properties of known indicators, and, in particular, research indicators are subject to

strong effects of non-linearity (Egghe, 2005; van Raan, 2004). It has been known for a long time, for example, that most bibliometric indicators are not size-independent (Katz, 2000, 2005), that the statistical properties at lower level of aggregation are not the same than at higher level (van Raan, 2006), or that the distributions obtained using averages of ratios are different from those obtained using ratios of averages (Larivière & Gingras, 2011). With respect to composite indicators, other methodological caveats must be observed (OECD, 2008). This problem is adequately addressed by the ontology-based data management approach we advocate here. Following this approach, the open-data platform is organized in such a way that the concepts and relations are logically connected with an explicit meaning and structure. In this way, the construction of new indicators benefits from the formalization of the logical relation among the constituent elements and drives the examination of their statistical properties. These properties cannot be derived logically, but must be examined empirically from the statistical properties of the constituent elements. The ontology-based approach makes the kind of empirical analysis needed explicit, in order to highlight the statistical properties of the new indicators. Thus, to make an example, if a new indicator is generated in the form “number of academic patents per thousand euros of research funding,” its statistical properties derive empirically from the underlying distributions of patents and funding; therefore, they cannot be deduced logically. However, the ontology-based approach dictates the procedure by which the properties of the constituent elements are empirically combined.

In the next section, an example of how to construct indicators with the ontology is reported.

An Example of Construction of Indicators

Let us illustrate an example of how the ontology might work. There are concepts that are described by a vector of attributes and are linked among them by relationships. There are formal rules by which it is possible to verify whether the description of attributes and relationships satisfies logical properties. This is the powerful contribution of theoretical computer science to the field of construction of indicators. If attributes and relationships are formally correct, then concepts are completely described by attributes, and relationships can be used for ascribing new attributes to concepts. In addition, it is possible to generate new relationships from existing ones, based on the joint ownership of attributes.

Let us take the example of a simple scientific research system in which the scholars are affiliated with universities; they publish articles and (sometimes) invent patents while receiving grants from a funding agency. Let us ignore for the moment the non-university sector and assume that individuals can be identified (by name, surname). (This is not obviously true due to well-known issues of ambiguity.) There are four logically independent concepts referring to individuals: (a) scholars are individuals affiliated with universities; (b) authors are individuals whose names and surnames appear on a publication; (c) inventors are individuals

TABLE 3. A sample of concepts, attributes and relationships in a sketch of a simple research system.

Concept	Selected attributes	Notation	Relationship
Scholar	Name Surname Affiliation (...)	Scholar: S_i $i = 1, 2, \dots, S$ (count) S = total scholars in the reference set	Affiliation relation (scholar $\rightarrow$ university)
University	University name City (...)	University: U_i $i = 1, 2, \dots, U$ (count) U = total universities in the reference set	
Publication	Title Journal Author Affiliation of author Acknowledgment (...)	Publication: P_i $i = 1, 2, \dots, P$ (count) P = total publications of the reference set Acknowledgment: K_i $i = 1, 2, \dots, K$ (count) K = total acknowledgments of the reference set	Authorship relation (scholar $\rightarrow$ author $\rightarrow$ publication) Affiliation relation (author affiliation $\rightarrow$)
Author	Name Surname Publication	Author: T_i $i = 1, 2, \dots, T$ (count) T = total authors in the reference set	
Patent	Inventor Assignee Title (...)	Patent: R_i $i = 1, 2, \dots, R$ (count) R = total patents in the reference set	
Inventor	Name Surname Address City (...)	Inventor: I_i $i = 1, 2, \dots, I$ (count) I = total inventors in the reference set Author-inventor: Y_i $i = 1, 2, \dots, Y$ (count) Y = total author-inventors in the reference set	Inventorship relation ((name, surname) $\rightarrow$ inventor $\rightarrow$ patent) Author-inventor relation ((name, surname) $\rightarrow$ scholarinventor)
Funding agency	Agency name Programme Grant Principal investigator	Agency: A_i $i = 1, 2, \dots, A$ (count) A = total agencies in the reference set Grant: G_i $i = 1, 2, \dots, G$ (value) G = total amount of grants of the reference set	Funding relation ((name, surname) $\rightarrow$ principal investigator $\rightarrow$ funding agency) Acknowledgment relation (author publication $\rightarrow$ acknowledgment $\rightarrow$ funding agency $\rightarrow$ grant)
Principal investigator	Name Surname Grant	Principal investigator: PI_i $i = 1, 2, \dots, J$ (count) PI = total principal investigators in the reference set	

whose names and surnames appear on a patent; and (d) principal investigators are individuals who receive a grant from an agency. There are also two concepts referring to documents (publications and patents) and two concepts referring to organizations (universities and funding agencies). Table 3 offers a highly simplified sketch of these elements and is presented as an illustrative example.

There are also a few fundamental relations, such as authorship, invention, affiliation, funding and acknowledgment. Using these relationships, it is possible to identify attributes that are jointly owned by separate concepts. For example, the same name (surname); that is, the same individual or physical person, may be at the same time a scholar, an author, an inventor and a principal investigator. If this is the case, all attributes logically ascribed to one of the concepts can be ascribed to the others (i.e., it is possible to identify author-inventors, or to examine how grants relate to the joint production of patents and publications). The relationships must cover the logically relevant situations, so they must admit cases of non-application. In most cases, for example, authors will not be inventors. In some cases, scholars will not be authors (i.e., non-active researchers), or authors will not be scholars (i.e.,

authors not affiliated with a university). The generative power of the ontology is another powerful contribution of theoretical computer science to the field of construction of indicators.

It should be noted that once the concepts are fully described by attributes and the relations are fully spelled out, then a vast potential is opened for logically correct inferences. Note also that, once this work is done, the adoption of international standards takes a tremendous impact, since the certainty of the information is propagated across the entire system. Thus, once the ID of publications (DOI), authors (author ID), universities, PROs, and funding agencies are subject to a common standard, and are then drawn from their respective Master File, the potential for automatic extraction of data and construction of indicators becomes enormous. Let us offer a simplified illustration of the construction of indicators.

Suppose we are interested in the impact of the funding of an agency and we define an indicator of number of publications per euro (or dollar) spent. This indicator is not already available in the data. Under a traditional approach, the funding agency should commission a study that asks those who receive a grant from the agency to declare which publications they have produced. Alternatively, the agency might simply

TABLE 4. Examples of construction of indicators.

Suggested name of indicator	Description	Building blocks of the indicator	Formula of the indicator
Scientific impact of funding for Agency “A” (SIF ^A)	Number of publications per dollar spent (at agency level)	P_s^A = Number of publications of scholars with acknowledgment for agency A P^A = Number of publications of all scholars in the reference set with acknowledgment for agency A G_s^A = Grants assigned by agency A to scholars G_{PI}^A = Grants assigned by agency A to principal investigator PI G^A = Grants assigned by agency A to all principal investigators of the reference set Note: S is the subsample of PI which are authors of publications with acknowledgment for agency A.	$SIF^A = P^A/G^A$
Scientific impact of funding for all agencies (SIF)	Number of publications per dollar spent (at national level)	P = Number of publications of all scholars with acknowledgment for all agencies (at national level) G = Total amount of grants assigned by all agencies to all principal investigators (at national level)	$SIF = P/G$
Technological impact of funding for Agency “A” (TIF ^A)	Number of patents per dollar spent (at agency level)	R_y^A = Number of patents of author-inventors that are also principal investigators of Agency A R_y = Number of patents of all author-inventors (at national level) G_y^A = Grants assigned by agency A to all scholars who are also author-inventors	$TIF^A = R_y^A/G_y^A$
Technological impact of funding for all Agencies (TIF)	Number of patents per dollar spent (at national level)	R_y = Number of patents of author-inventors that are also principal investigators of all agencies (at national level) G_y = Total Grants assigned by all agencies to all scholars who are also author-inventors (at the national level)	$TIF = R_y/G_y$

ask grantees to use the acknowledgment as a mandatory condition to receive grants (this is the approach of several funding agencies, such as the European Research Council). If this is the case, then it is possible to define an acknowledgment relation, linking the same individual (name, surname) to two different concepts: author and principal investigator (grantee). It is then possible to build up an impact indicator, first at the level of a single agency, then at a national level (provided a Master List of funding agencies is established).

Suppose further that the agency wants to understand whether grantees are also producing inventions out of the money they receive. In other words, there might be a question of technological impact of funded research – a highly relevant policy issue nowadays. Again, in a traditional approach, there would be the need for a commissioned study. In our approach, it is possible to combine two relationships (authorship and invention) and obtain a new concept, author–inventor, which can be predicated of an individual and identified (name, surname). If this is the case, we can ascribe patents to author–inventors, and ascribe grants to them based on the acknowledgment relationship. It is then easy to build up a new family of indicators. See Table 4 for some illustrative examples. Note that in this way it is possible to locate all patents of authors, and not just those that are officially assigned to the organization of the authors.

Building the Data Infrastructure

In this section, we describe a data infrastructure that might be used to check whether existing data permit the implementation of the methodology illustrated in the previous section. The data infrastructure allows the construction of new indicators that can be used not only in isolation, but

also in the construction of new ranking indicators based on advanced input–output modelling techniques rather than on composite indicators.

The Smart.CIEU (Sapienza Microdata Architecture for education, Research and Technology studies – a Competence-based data Infrastructure on European Universities) is a general conceptualization of a data infrastructure created within a research project funded by Sapienza University of Rome and owned by the Department of Computer, Control and Management Engineering Antonio Ruberti, Sapienza University of Rome. The design of Smart.CIEU has been made possible by the integration of several data sources. This is entirely based on data on European higher education institutions, thanks to the integration of several projects.

Firstly, thanks to the pioneering Aquameth (Bonaccorsi & Daraio, 2007; Daraio et al., 2011), Eumida projects (Bonaccorsi, 2014), and the subsequent launch of the ETER project (European Tertiary Education Register, <http://eter.joanneum.at/imdas-eter/>), an official census of European higher education institutions has been established. An official list of institutions, validated by the National Statistical Authorities (compliant with national legislation) is now available. Each institution is characterized by a vector of descriptors, including the geographic location at the NUTS 2 level (region). Additional geo-referentiation at the NUTS 3 level (province) has been subsequently provided on Eumida data (Varga & Horvath, 2014). The Eumida project made a data set limited to basic descriptors of universities (Data Collection 1) publicly available, while Data Collection 2 – related to research active institutions – was not published. The ETER project updated the Eumida collection for 2011 and 2012 (a new ETER project is collecting data for

2013 and 2014) and has negotiated the conditions for the publication of data with the National Statistical Authorities. Thus, a public data set on the census of higher education institutions is now available.

The comparability of the data is made possible by the fact that the National Statistical Authorities follow the same statistical Handbook (UOE, 2006). In addition, the precise interpretation of definitions, and the problems arising from national reporting systems and standards, have been the object of the data quality approach that was developed within the project, which leads to intense discussion and clarification. The data quality approach will be consolidated and a recommendation will be provided within the data quality report, which will be made available together with the data. Thus, in the ETER data set the main comparability problems across countries have been described and discussed at length. They include detailed metadata, offering careful descriptions of the way in which national data are collected.

Secondly, the location of institutions can be exploited by associating a number of indicators with the institutions located in the same province (where NUTS 3 data are available) or region (where only NUTS 2 data are available). By using Eurostat official data, the economic, social, cultural, and demographic context of institutions can be fully characterized. Contextual or environmental variables also can be used to construct indicators of the impact of universities on society.

Thirdly, by using the official name of institutions, it is possible to allocate research output to institutional affiliations. We exploited the Global Research Benchmarking System (GRBS) project, which allocated Scopus publications to individual universities in East Asia, North America, and Europe. With respect to European universities, the affiliations in Scopus have been cleaned by the same group of experts that worked on the Eumida project. These data are then compared with SCImago data, which are also based on Scopus. However, the GRBS data are more fine-grained, since they have been made available for 251 separate subject categories. These data were available on a complimentary basis and only for a window of time (2007–2010), which nicely overlapped the Eumida data set (2008–2009). The use of the same official census of universities in both the descriptors and the publication data leaves no uncertainty on the attribution of publications to universities.

Thus, the integration of data for the period 2007–2010 was highly reliable and still offers a test of the feasibility of integration on a permanent basis even past 2015. However, a problem that should not be underestimated relates to the quality of Scopus data on publications when we increase their granularity. In many cases, it is problematic to quantify these problems, if they pertain to Scopus alone or are present also in other citation indices, as they relate to:

- The journal classification system that, at lower aggregation levels, does not always seem reliable;
- Inconsistencies in the assignment of source types versus document types;
- Publications that remain “in press” for years;

- Not very well structured author names (e.g., first names vs. initials);
- Author affiliations;
- Missing publications;
- Missing bibliographic information (e.g., cited references);
- Duplicate publications (see Valderrama-Zurián, Aguilar-Moya, Melero-Fuentes, & Aleixandre-Benavent, 2015);
- Wrong DOIs.

Given these problems, it could be useful to invest in an open access citation index, and in interlinked publication and data repositories as alternatives to commercial databases. These could develop into a publicly available publication index, which is fully integrated into the data infrastructure.

By integrating these three data sources, it is possible to obtain the basic indicators listed in Table 5.

We have to admit that the data on R&D expenditures, patent applications, spin-offs, and categories of expenditures are far from being complete. This is because most national statistical authorities did not provide them, nor did they allow the relative information to be published online.

Nevertheless, the other indicators have been tested for reliability in the studies described earlier. They maintain the same statistical properties of constituent data. This means that in the future an open-linked data infrastructure might include a menu of operations over basic indicators, leading to the construction of new indicators. In that way, users might be allowed to build up custom-based indicators without having to start from scratch.

The crucial point here is that these projects have been conceived with a view to the integration of new data. The potential is already visible in two directions: (a) integrating covariates using the university as the unit of observation; and (b) integrating covariates using the territory as the unit of observation.

The developments in (a) are made possible by the fact that by building up the census of HEIs it has been possible to reconcile the data on students (which are in the official domain of higher education statistics) with the data on research (which are in the official domain of S&T statistics). Although the data on research input, such as the R&D expenditure (OECD-Eurostat), are not yet available at a microlevel (university level) due to the low reliability of the allocation of higher education expenditure between research and teaching, it has been possible to integrate the data on research outputs, such as doctorate degrees and scientific publications. With respect to (b), the georeferentiation of HEIs offers the possibility of not only integrating the data available in the census with a host of other data with a territorial dimension at NUTS 2, but also, with limitations, integrating the data at the NUTS 3 level.

The potential for this approach is already visible in a number of recent developments. In Bonaccorsi (2014), several authors combined Eumida data with covariates of various kinds, using the university or the territory as the unit of observation (e.g., publications, academic patents, creation of new technology-based firms at a province level, performance of research teams at a university level), showing the potential for empirical research.

TABLE 5. State of the art of integration of official statistical data on higher education institutions with output data and territorial covariates.

Identifiers	Institutional code Name of the institution (in own language) Name of the institution (in English)
Basic institutional descriptors	Country Legal status Type of institution, national definition Type of institution standardized Foundation year Current status year University hospital (dummy)
Geographic information	Region of establishment, NUTS 2 code Province of establishment, NUTS 3 code
Educational activities	Highest degree delivered Number of enrolled students at ISCED levels 5 • By field of education • By national/foreign origin Number of graduates at ISCED levels 5 • By field of education • By type of degree (diploma, bachelor, master, long cycle) • By national/foreign origin Distance education institution (dummy) Research active institution (dummy) Number of enrolled students ISCED 6 • By field of education (FoE) • By national/foreign origin Number of graduates ISCED 6 (doctorates) • By field of education (FoE) • By national/foreign origin <i>R&D expenditures</i> <i>Patent applications</i> <i>Spin-off companies</i>
Expenditure	Personnel expenditure Non-personnel expenditure Capital expenditure
Revenues	Core budget Third party funding Tuition fees
Personnel	Total Funding from Private Sector Number of academic personnel in full time equivalent (FTE) and headcount (HC) • By field of science (FoS) • By national/foreign origin Number of administrative personnel in FTE/HC
Scientific publications ^b	Number of institution publication output International collaborations: share of publications produced in collaboration with foreign institutions High quality publications: as a share published in the most influential scholarly journals of the world (top 25%) Normalized impact: relationship of an institution's average scientific impact (average citations) and the world average, which is 1 Specialization index: extent of thematic concentration/dispersion of an institution's scientific output (0 for generalistic to 1 for specialized) Excellence rate: % of an institution's output included into the set formed by the 10% of the most cited articles in their respective scientific fields
Scientific publications by subject area ($n = 250$) ^c	Number of publications by subject area Percentage publications in top source titles • Top 10% journals • Top 25% journals Number of citations Percentage of citations from top source titles • Top 10% journals • Top 25% journals

TABLE 5. *Continued*

Regional covariates ^d <i>Eurostat regional data at NUTS 2 level. Selected indicators</i>	H-index Regional gross domestic product (PPS per inhabitant) Employment rate of the age group 15-64 Total intramural R&D expenditure (GERD) Millions of PPS Total R&D personnel and researchers - head count Patent applications to the EPO by priority year – Number Stud ISCED5_6 as % of pop 20-24 years 2008
Local covariates ^e <i>Eurostat regional data at NUTS 3 level. Selected indicators</i>	Total population (possibly by age group) and population density Total GDP in PPS and GDP per capita Gross value added at basic prices, possibly by NACE sector (11 sectors) in G€ Employment (in 1,000 persons) possibly by NACE sector (11 sectors)

Note. ^aItems in italics are available for a subset of countries (Eumida Data collection 2).

^bSource: SCImago based on Scopus data.

^cSource: Global Research Benchmarking based on Scopus data.

^dPossibility to combine with a larger set of regional indicators at NUTS 2 level available on Regions and Cities Eurostat data set.

^ePossibility to explore also Eurostat data on metropolitan regions and Urban Audit indicators on selected cities.

With respect to integrating data at the level of universities, the Eumida data have been integrated with data on publications, and Daraio et al. (2015b) carried out a directional distance analysis of efficiency, economies of scale and specialization. In contrast, Daraio et al. (2015a) propose a conditional multidimensional ranking, using the same data.

With respect to integrating data at a regional level, Vertesy, Annoni, and Nardo (2013) have integrated the Eumida data with a number of indicators at a regional level.

As shown in Table 6, in future studies it would be possible to integrate other sources of information already available, again following either the direction of integration assuming the university as unit of observation (European funding, U-Multirank, ERA survey), or the territory (Regional Innovation Scoreboard). In some cases, access to these data is not public.

Addressing User Requirements and Shifting the Existing Paradigm

Needless to say, the logic of indicators does not follow the logic of policy makers or stakeholders. Indicators must satisfy statistical properties and be based on sound databases. They are produced at fixed dates with a long and rigid preparation process. In contrast, decision makers have a large variety of information needs, and, above all, want to satisfy these needs at short notice.

In principle, indicators should be designed according to the needs of the users. However, the needs of the users are highly fragmented and idiosyncratic. Universities are multi-input/multi-output organizations, with a large number of heterogeneous stakeholders and with highly differentiated production technologies. Indicators must follow the informational needs of sophisticated and heterogeneous users closely.

In order to design a data platform with a conceptual framework able to account for a very broad user require-

ments perspective, we refer to the work of a group of experts (Expert Group on Assessment of University-Based Research; AUBR) of the European Commission (EC). Table 7 has been developed by the AUBR group of experts and gives an overview of the sheer variety of the types of needs that indicators should address. The expert group was asked by the EC to address the question: What are the criteria for the assessment of university-based research? Instead of producing a unique list, the group suggested many different lists of indicators, each of which was tailored to the needs of a particular audience or group of stakeholders. The main users groups considered were:

- Higher education (HE) management and governance
 - Governing bodies/councils
 - HE executives/management
 - HE research groups
- Governments and government agencies
 - European Union and national governments
 - Ministries of education/higher education or enterprise and employment
 - Local and regional governments
 - HE agencies
 - Other government agencies
- Academic organizations and academies
- Individuals
 - Academics and researchers
 - Students
- Peer HEIs
- Industry partner organizations
 - Private firms and entrepreneurs
 - Public organizations
 - Employers
- Civic society and civic organizations
- Ministries of higher education in developing countries
- Sponsors and private investors
 - Benefactors/philanthropists
 - Alumni
- Public opinion.

TABLE 6. Feasibility of further integration in the open platform.

European funding ^a	EU Framework programmes • Participants • Funding • Role (e.g., coordinator)
Regional Innovation indicators ^b	Sales of new to market and new to firm innovations Employment in knowledge intensive activities SMEs introducing marketing or organizational innovations SMEs introducing product or process innovations EPO patent applications Innovative SMEs collaborating with others SMEs innovating in-house Non R&D innovation expenditures R&D expenditure in the business sector Population with tertiary education
U- Multirank ^c	Indicators of teaching and learning Student-staff-ratio Graduation rate Percentage of academic staff with PhD Percentage of graduates graduating in norm period Rate of graduate unemployment Inclusion of work experience Indicators from the student survey • Overall learning experience • Quality of courses & teaching • Organization of program • Contact to teachers • Social climate • Facilities ○ libraries, ○ laboratories, ○ rooms, ○ IT • Research orientation of teaching/programme • Inclusion of work experience/practical elements Indicators of research External research income (per FTE academic staff) Doctorate productivity Total publication output (per FTE academic staff) Field-normalized citation rate Highly cited research publications Research orientation of teaching (student survey) Indicators of knowledge transfer Income from private enterprises (research contracts, service contracts, licenses, etc.,) Joint publications with industry Patents (per FTE academic staff) Co-patenting with industry per FTE academic staff Indicators of international orientation International orientation of programmes (rating indicator) Opportunities to study abroad (student survey) Percentage of international academic staff Percentage of PhDs by foreign students International joint publications International research income Indicators of regional engagement Percentage of graduates working in the region Student internships in local enterprises Degree theses in cooperation with local industry

TABLE 6. *Continued*

European Research Area indicators ^d	Regional joint publications
	Income from regional sources
	Large array of questionnaire-based data from a large sample of European research performing and research funding organizations

Note. ^aInformation on EU programmes funding is publicly available in a database format at http://cordis.europa.eu/projects/home_it.html. Data have been organized by organization in a restricted access website, see https://webgate.ec.europa.eu/e-corda/resources/pdf/BO_XI_Quick_user_guide.pdf. Data are available by subject.

^bhttp://ec.europa.eu/news/pdf/2014_regional_union_scoreboard_en.pdf. Data are available only at regional level (NUTS 2).

^c<http://www.u-multirank.eu/#!/home?trackType=home§ion=entrance>. Data are available for a non-representative sample of universities in Europe and other areas.

^dSee the questionnaires for Research Performing Organizations (RPOs) and Research Funding Organizations (RFOS) and the associated Glossary. http://ec.europa.eu/research/era/pdf/ERA_Survey_2014_Glossary_v6.pdf.

Moreover, these lists of indicators were produced without any consideration of the current availability of data, but only from the point of view of the *desiderata* of users. Given this premise, it is not surprising that many of the listed indicators are not currently available – not at a cross-country comparable level, nor even at a country level (in most cases).

The last column of Table 7 offers a preliminary assessment of the feasibility of deriving these indicators from the platform described earlier. It is clear that, while our proposal does not solve all the information needs, it contributes to a significant improvement over the current state of the art.

Thus, the previous report is a confirmation that, when asked about their needs for indicators, users express a clear preference for granularity (in this case, institutional and disciplinary) and cross-referencing (in this case, among various indicators of inputs and outputs). This creates a difficult dilemma between the availability of indicators and the cost of producing them.

We suggest that, faced with this dilemma, there are two lines of action. One line is the traditional “silos” approach. Faced with a need for indicators, policy makers commission the construction of a new database, or the integration of several existing databases. Inevitably, the resulting database is not inter-operable with other databases, and is not scalable. It serves, perhaps optimally, the current information need, but cannot address (even slightly) different needs, or the same needs in the future, unless substantial money is spent on completely revamping the current database.

The other line of action is based on the construction of a platform to explore the power of open-linked data. Unavoidably, the construction of a platform requires an initial investment. This is currently greatly reduced by the ongoing process of standardization, described earlier, which implies that interoperability is not built from scratch. Nevertheless, it cannot be denied that an initial cost must be absorbed by some public or non-profit institution, or coalitions thereof.

Following a real options approach in investment theory (Li & Johnson, 2002), we can conceive systems of indicators as an investment activity of government and public administration. The data platforms are then considered as assets allowing repeated use. In this context, investment costs are made by up-front costs for the platform, maintenance costs,

and recurring costs for projects. Instead, the revenues are the gains from better decisions in policy making; the possible use for performance-based allocation of public resources; the possible use for strategic priorities in S&T; or to set up public subsidies to firms for industrial R&D.

The traditional approach to investment appraisal based on the comparison of up-front costs at the initial time period with the discounted cash flow from revenues at subsequent periods of time could be overcome by adopting a real options approach (Li & Johnson, 2002), which, in the appraisal of investment, includes the options that have been made possible by the investment. Interestingly, within this framework there are learning and transfer options.

Therefore, we must shift the paradigm:

- From proprietary sources to open-linked data sources, including explorations and investments in the open access citation index, interlinked publication and data repositories as alternatives to commercial-based products, which could develop into a publicly available publication index;
- From ad hoc indicators to indicators that can be generated reliably by a combination of existing data in a platform that was designed considering a wide range of users’ requirements, as illustrated in the previous section;
- From indicators mainly based on research to a comprehensive set of indicators, including several outputs, and taking into account the structure of the input.

Conclusions

In this article, we aim to introduce a change in the way indicators are designed. A departure from the traditional approach is suggested:

- Start top–down with a thorough conceptual analysis of the research and the higher education system;
- Draw all linkages between concepts, particularly the sharing of concepts between entities;
- Examine the linkages down to the lowest possible level of granularity, in order to ensure that the sharing is complete;
- Generate new indicators through the combination of shared concepts;
- Implement the new indicators with existing data, or invest into the areas that may produce the largest increase in connectivity among concepts.

TABLE 7. User requirements for indicators on European higher education.

Indicator	Description	Notes	Feasibility
RESEARCH PRODUCTIVITY			
Research publications and outputs	A count of publications and other research outputs.	Depending on purpose only selected types of publications can be counted. Publishing is vital for progress in science scholarship. Different disciplines produce different types of research outputs. Emphasis on quantity of publication. Suitable databases for a variety of disciplines and research related outputs, especially in social sciences and humanities.	Yes
Research outputs per “Research Academic” staff	Number of publications and other research outputs per academic staff or full-time equivalent (FTE).	Supports cross-institutional comparisons, adjusted for scale of institution. Comparable definition of “Academic Staff” and ‘Research Time’ can be difficult. Agreement on definition of “Research Academic.”	Yes
QUALITY AND SCHOLARLY IMPACT			
Number and percentage of publications in top-ranked, high impact journals	The number or percentage of journal articles published in the top-ranked, high impact journals for the fields of research.	Widely used, especially in the exact sciences which tend to be well covered. Data must be accurate and verified. Although one of the most popular indicators, it is not always the most appropriate one. Especially in social sciences and humanities, expert rankings do not correlate very well with impact factors. In these fields and in engineering, other sources are important as well (books, proceedings). Discipline specific journal rankings, especially in social sciences and humanities, based on expert opinion in combination with indicators. Value of developing a ranking or hierarchy of scientific-scholarly publications.	Yes
Citations	Citation data are derived from citation indexes, i.e., databases that do not only contain metadata	In the exact sciences, peers tend to consider citation impact a relevant aspect in assessments of research performance. Citations reflect intellectual influence but do not fully coincide with research quality. Expansion of existing databases and creation of new databases (e.g., based on data from institutional repositories) will include publications but also their reference lists. Principal indexes are the Web of Science, Scopus, and Google Scholar. Widely used, especially in the exact sciences which tend to be well covered, although the most popular indicators are not always the most appropriate ones. Data must be accurate and verified. Are of limited value in disciplines not well covered by the citation indexes, especially certain parts of social sciences, humanities, and engineering. Improve the value of this indicator and coverage of disciplines. Theoretical research into the meaning of citations (clusters) in social sciences and humanities.	Yes
Number Keynote Addresses at National and International Conferences	A count of the number of invited and keynote addresses given at national and international conferences	Used as proxy for quality, impact and peer esteem. Data can be verifiable by conference programme. No agreed equivalences that apply internationally and facilitate comparison across disciplines. This will probably require direct entry by researchers. A list of internationally comparable items for different disciplines might help a lot.	No
Number Prestigious National/International Awards and Prizes	A count of the number of prestigious national and international prizes	Used as an indicator of research quality and impact. Data are verifiable. No agreed equivalences that apply internationally and facilitate comparison across disciplines.	No

TABLE 7. *Continued*

Indicator	Description	Notes	Feasibility
	won either in total or per academic staff.	Unless lists are publically available this will require direct entry by researchers. A list of internationally comparable items for different disciplines might help a lot.	
International Visiting Research Appointments	A count of the number of visiting appointment at other academic and/or non-academic agencies and organizations.	Visiting appointments provide indication of peer esteem or support by the academic community. Numbers are verifiable. No agreed equivalences that apply internationally and facilitate comparison across disciplines. Will probably require direct entry by researchers.	No
Editorial and Refereeing for Prestigious National/International journals/publishers	A count of the number of national and international appointments as editor, member of editorial board or reviewer.	An indicator of the extent to which the researcher's opinion is highly regarded by the academic community. Data are verifiable. No agreed equivalences that apply internationally and facilitate comparison across disciplines. Unless lists are publically available this will require direct entry by researchers. A list of internationally comparable items for different disciplines might help a lot.	No
INNOVATION AND SOCIAL BENEFIT			
External research income	Level of funding attracted by researchers and universities from external sources, including competitive grants and research income from government, industry, business and community organizations.	Comparable data, verifiable through audit, are useful for comparing research performance across the system and within universities. Willingness of industry to pay for research is a useful indicator of its anticipated contribution to innovation and the economy. Levels of external funding vary greatly across disciplines. For example, in countries where over half the total pool of funding is allocated to medical research, universities that do not have medical faculties will inevitably secure less funding than those with medical faculties. Data collection may be difficult in case of funding by end users because this information is not known to the university administration. Agree international comparative database.	Yes
Number and percentage competitive grants won	Level of funding won competitively –this is a subset of the indicator above.	Comparable data, verifiable through audit, are useful for comparing research performance across the system and within universities. Levels of external funding vary greatly across disciplines. Agree international comparative database.	No
Research income per academic staff or FTE	Research income per academic staff or FTE	Supports cross-institutional comparisons, adjusted for scale of institution. Important measure of research activity. Comparability is dependent upon institutional mission, context, and discipline. Data need to be adjusted to scale and mission of university.	No
Employability of PhD graduates	Industry employment of PhD graduates	Industry employment of PhD graduates can be an indicator of the contribution of research to society. Used to measure the quality of the graduates, and impact of research on teaching. Career paths and opportunities can differ for different disciplines. Employability can be sensitive to other factors, such as the regional or national economy. Important to develop methods to track graduate employability and career paths. Harmonize the stage(s) post-graduation at educated & skilled workforce.	No
Commercialization of research generated intellectual property (IP)	Provides measure of the extent of income from commercialization of intellectual property	This is an area of increasing significance to policy makers. Indicator is an important link between IP, commercialization and economic benefits.	Yes (̂)

TABLE 7. *Continued*

Indicator	Description	Notes	Feasibility
	created through patents, licences or start ups.	Patents are a very poor indicator of commercialization. They are sensitive to national context – and to discipline. Databank on university-related inventions should be developed.	
End User Esteem	Includes policy, technical or commissioned reports; consultancy and external contracts; architectural or design awards; etc.	Willingness of external stakeholders to use and/or pay for research is a useful indicator of its anticipated contribution to innovation and the economy. Different opportunities for different disciplines. Lack of agreed basis of capturing data and comparability could undermine legitimacy. Agree basis of international comparability and verifiability.	No
Number and percentage funding from end users	Provides measure of the extent of income from external-commissioned or contracted work (industry, professions, government, community)	This is an area of increasing significance to policy makers. Indicator is an important link between research and social and economic benefits. Different opportunities for different disciplines. Lack of agreed basis of capturing data and comparability could undermine legitimacy. Agree basis of international comparability and verifiability.	Yes
SUSTAINABILITY AND SCALE			
Post graduate Research Student Load	The ratio of research students (or PhD students) per academic staff or per “Research Active” staff	Key indicator of research intensity, indicating the scale of the research enterprise. Practices differ across disciplines – large research teams are a common feature of the bio- and medical sciences. Agree basis of international comparability and verifiability.	Yes
Involvement of early career researchers in teams	Number or percentage of early stage researchers involved in research	An indicator of research intensity, the scale of the research enterprise, and future activity. Practices differ across disciplines – large research teams are a common feature of the bio- and medical sciences. Agree definition of “early career researcher,” and basis of international activity.	No
Number of collaborations and partnerships	A count of national and international collaboration with other universities and/or with public-private and NGOs, etc.	Because research is increasingly conducted in collaborative teams, nationally and internationally, this is an important indicator of research involvement and scale of activity. Can be difficult to capture and verify the data due to lack of clarity as to what is being measured. Agree precise definition, inter alia: university-university, university-external stakeholder, national, European or international.	No (*)
Doctoral Completions	The number PhD and equivalent research doctorates and, as appropriate, research masters degree completions.	Data are verifiable by universities although there can be a time lag. Rates of completion may differ across disciplines. Different disciplines may prioritize masters and PhD activity. Require common standard for doctorates.	Yes
RESEARCH INFRASTRUCTURE			
Research active academics	Number or equivalent full-time (FTE) of “research active” academics employed by a university.	“Research active” is established by setting threshold levels of performance for a specific period. Important indicator of research capability. No clear definition of “Research Active.” Common definition and international comparability of “Research Active.”	Yes
Percentage “Research Active” per total academic staff	Ratio of the number of “Research Active” per total academic staff.	Indicator of research intensity. No clear definition of “Research Active.” Common definition and international comparability of “Research Active.”	Yes
Total R&D investment	Total investment in university-based R&D	Strong predictor of research performance.	No

TABLE 7. *Continued*

Indicator	Description	Notes	Feasibility
	from all sources, including External research income and university resourcing of research	Difficult to get valid, comparable institutional data, even within the same institution. Agree basis on which to calculate full cost of research investment, salaries, and overheads. Can be difficult to fully calculate university resourcing of research.	
Research Infrastructure and Facilities	Number of research laboratories, books in the library, and/or electronic journal access, supercomputing access	Information provided at the level of the institution. Difficult to get valid, comparable data. Favors older, well-endowed universities. Develop appropriate comparative indicators.	No
Research Ethics	Comprehensive process ensuring good ethical practice is promoted and promulgated.	Important measure of research rigor and integrity, and the effect and purpose of the research. Peer review would be most useful. Ethical statements regarding the use of research and the source of research funding, e.g., tobacco or armaments, can be very controversial. Develop appropriate indicators to ensure good ethical practice is promoted without interfering in processes of discovery.	Yes

Source: Our elaboration on European Commission (2010).

Note. By “Yes” we mean the potential availability of data covering all institutions, excluding the cases in which data are available on a sample (e.g., ERA survey or U-Multirank).

Following this approach, once indicators are created, their feasibility is immediately visible. If there is an area of the infrastructure uncovered by existing data, it is clear where to invest in the production of new data.

Furthermore, the process is recursive. Once new indicators have been generated, they will share the concepts that have originated them. Then they enter, again, into the process of identification of shared concepts. Indicators built over other indicators become feasible.

An open-data platform approach, which we advocate here, encourages and supports new research developments in the generation of new indicators carried out by scientists, which exploit the accessibility and transparency of data. In this way, there will be opportunities for the creation of new indicators beyond the short-term needs of policy makers.

The open-data approach, offering full documentation on data and explicit articulation of logical linkages, makes obsolete the traditional training and accreditation approach, in which users of indicators were dependent on the training provided by the owners of the data. Communities of users can, in fact, contribute to the improvement of the documentation and identify pitfalls and shortcomings of indicators.

Recently, a call for a European integrated research information e-infrastructure also has been supported from a report of the European Parliament: “The trends in the Member States and their development or national research information systems have created a momentum for the development of a pan-European comprehensive research information infrastructure, geared to supporting science management as well as serving scientists. European policy-makers should exploit this opportunity and support and coordinate the development of a European integrated research information system” (EPRS, 2014, p. 14).

We claim that an *intelligent* integration of existing data in an open-linked data platform may permit the construction of new indicators, without having to design the indicators on a custom basis.

The integration of existing heterogeneous data in the frame of an ontology-based data management approach, which we advocate here, could be the basis for the development of a full-scale ontology to produce new indicators on higher education and research, according to a broad set of users’ requirements.

The realization of a sustainable infrastructure, based on such an ontology, could follow different business models. It could be based on:

- The direct involvement of the EC, with a scientific board to coordinate the activities and its maintenance, while the management of the system could be carried out in-house (e.g., through the Joint Research Centre) or be contracted out;
- A public-private sponsorship model, which is based on a collection of private foundations and public funding agencies. Also in this model, the creation of a scientific board should be provided and the management of the system could be carried out by a non-profit organization (see, e.g., the management of ORCID);
- A completely decentralized, individual-based sponsorship model – a kind of Science 2.0 model – based on the collection of donations from individuals, private foundations, and public funding agencies. A model that relies on the mobilization of researchers, on crowd-funding (see, e.g., Wikipedia, Research Gate).

We argue that this approach might help to address a number of unsolved policy issues. We have only picked up a few examples here. The examples we will report here are also

useful to show the main difference of our approach with respect to other projects that have been proposed in the field of research evaluation; in particular with respect to the STAR-METRICS initiatives developed in the United States and recalled in the introductory section of this article. Our approach is broader, as described in the previous section, and requires the development of a rich and articulated ontology-based data integration model. On the contrary, the STAR-METRICS initiative is focused on governmental funding and the tracking of their scientific and technological impact.

Firstly, a large area of policy concern refers to the *funding of higher education*. In almost all countries, a per capita funding scheme has been adopted using student numbers, while in some countries, an additional performance-based scheme is used. Governments often ask whether public money is used efficiently with respect to other countries. However, it is extremely difficult to carry out input–output and efficiency analyses, or even to compare the cost per student or other simple cost indicators across countries, due to the lack of reliable and comparable integrated data on research and education.

Secondly, another area of hot policy debate is the *position of universities in international rankings*. Particularly in Europe, it is often asked why universities do not show up in the top positions, while resources are poured into them. Sometimes the concentration of resources in a few universities is recommended, as well as a process of consolidation via mergers or federations. However, these policy recommendations do not have strong empirical support, mainly because the data needed to examine the differential effect of selectivity and concentration of funding are not available. Again, following the proposed approach, it would be possible to compare individual universities controlling for country-level, size, and subject-mix effects.

Thirdly, a different area of policy attention refers to the *relationship between research, innovation, productivity, and growth*. This is a complex issue that does not admit a simple answer. One way to address this issue would be to examine microdata on research institutions. However, currently available data do not allow any meaningful integration of different outputs at the level of universities and PROs (e.g., publications, patents, spin-off companies, third-party research). Another way to tackle the problem might be to compare countries, regions or territories among themselves. However, data on research activities at a regional level, even more so at a local level, are not available or are highly aggregated. Since the impact depends on, among other things, the density of research activities and on disciplines, these policy concerns remain unsolved.

What might be the situation with an open-linked data infrastructure? Basically, it would allow the construction of custom-built indicators in a seamless way. Instead of being built upon queries on the data structure of existing databases, indicators should be built first in an abstract space of concepts. Concepts must have an explicit definition, which is independent from the particular database through which they can be observed. In this way, we break the dependency

of indicators on databases, which is at the origin of the traditional approach. Then, the logical structure of indicators might be expressed in terms of concepts and relations between them. Relations between concepts, in fact, are the backbone of custom-built indicators (e.g., publications and patents can be attributed to the *same* university; authors from academia and industry can be attributed to the *same* publication, etc.). If the concepts are well specified, the indicators are also uniquely defined. If the logical structure is well specified, the requirements on the underlying data will also be clear. Following our proposal, no data should be collected without a clear definition of the logical structure, but once the data are collected this way, the construction of new indicators is greatly facilitated. Furthermore, there is no need to use hierarchically structured databases, since the relations among concepts may also follow non-hierarchical patterns.

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